

24 Introduction to Bayesian Hierarchical Models

In a **Bayesian hierarchical**¹ model, the model for the data depends on certain parameters, and those parameters in turn depend on other parameters. Hierarchical modeling provides a framework for building complex and high-dimensional models from simple and low-dimensional building blocks.

Suppose we want to estimate the true, long run free throw percentage for a subset of NBA players². That is, we want to estimate each player's true probability of successfully scoring on any free throw attempt. We'll consider data from games in the 2022-2023 NBA season for players who were on the Philadelphia 76ers obtained from [basketball-reference](#) as of Feb 27, summarized in Section 24.1.1.

Example 24.1. Let θ_j denote player j 's true (long run) FT percentage; for example, θ_{Embiid} , θ_{Harden} , etc. Let n_j denote the number of free throw attempts and y_j the number of free throws made for player j .

1. One way to estimate θ_j is to pool all the data together. Over the 19 players in the data set, there are 1482 attempts, of which 1229 were successful, for an overall free throw percentage of 82.9%. The sample mean free throw percentage, among players with nonzero attempts, is 77.7%. So we could just estimate each θ_j with the overall free throw percentage (or the sample mean free throw percentage). What are some issues with this approach?
2. At the other extreme, we could estimate θ_j using just the attempts for player j . For example, use only the attempts for Embiid to estimate θ_{Embiid} , only the attempts for Harden to estimate θ_{Harden} , etc. What are some issues with this approach? (Hint: consider the players with no attempts, or few attempts.)

¹Hierarchical models are often called *multilevel*, *mixed effects*, or *random effects models*.

²Along with one bonus "player" who will illustrate some ideas.

3. In a Bayesian model, the θ_j 's would have a prior distribution. We could assume a Bayesian model with a prior in which all θ_j are independent. What are some issues with this approach?

Example 24.2. Let's first consider just a single player with true (long run) free throw percentage θ . Given n independent attempts, conditional on θ , we might assume that y follows a $\text{Binomial}(n, \theta)$ distribution.

1. If we imagine the player is randomly selected from among current NBA players, how might we interpret the prior distribution for θ ? Note: this question is mixing some Bayesian and frequentist ideas, but the purpose is to help intuition.
2. The prior distribution for θ has a prior mean; let's denote it μ . How might we interpret μ ?
3. The prior mean μ is also a parameter, so we could place a prior distribution on it. What would the prior distribution on μ represent?
4. The prior distribution for θ has a prior standard deviation; let's denote it σ . How might we interpret σ ?

5. The prior standard deviation σ is also a parameter, so we could place a prior distribution on it. What would the prior distribution on σ represent?
 6. In this model μ and θ are dependent. Why is this dependence useful? (Hint: consider another player.)
- A Bayesian hierarchical model consists of the following layers in the hierarchy.
 1. A model for the data y which depends on model parameters θ and determines the likelihood $f(y|\theta)$.
 - The model usually involves multiple parameters. Generically, θ represents a vector of parameters that show up in the likelihood function.
 - Remember: While the likelihood for a particular y sample is computed by evaluating the conditional density of y given θ , the likelihood function is *a function of θ , treating the data y as fixed.*
 2. A (joint) prior distribution for the model parameters θ , denoted $\pi(\theta|\phi)$, which depends on *hyperparameters* ϕ .
 3. A *hyperprior* distribution, i.e., a prior distribution for the hyperparameters, denoted $\pi(\phi)$. (Again, ϕ generically represents a vector of hyperparameters.)
 - The above specification implicitly assumes that the data distribution and likelihood function only depend on θ ; the hyperparameters only affect y through θ . That is, the general hierarchical model above assumes that the data y and hyperparameters ϕ are conditionally independent given the model parameters θ , so the likelihood satisfies

$$f(y|\theta, \phi) = f(y|\theta)$$

The joint prior distribution on parameters and hyperparameters is

$$\pi(\theta, \phi) = \pi(\theta|\phi)\pi(\phi)$$

- Given the observed data y , we find the joint posterior density of the parameters and hyperparameters, denoted $\pi(\theta, \phi|y)$, using Bayes rule: posterior $\propto$ likelihood $\times$ prior.

$$\begin{aligned}\pi(\theta, \phi|y) &\propto f(y|\theta, \phi)\pi(\theta, \phi) \\ &\propto f(y|\theta)\pi(\theta|\phi)\pi(\phi)\end{aligned}$$

- In Bayesian hierarchical models, there will generally be implied dependence between parameters. But dependence among parameters is actually useful for several reasons:
 - The dependence is natural and meaningful in the problem context.
 - If parameters are dependent, then all data can inform parameter estimates.

Example 24.3. Now suppose want to estimate the free throw percentages for J NBA players, given data consisting of the number of free throws attempted n_j and made y_j by player j . Specify a hierarchical model, including your choice of prior distribution. How many parameters are there in the model?

Example 24.4. Use software to fit the Bayesian model using the 76ers data. Summarize the posterior distribution of μ .

Example 24.5. For each player j , consider the posterior mean and posterior SD of θ_j .

1. What is the posterior mean of θ_j for players with no attempts? Why does this make sense?

2. Why is the posterior mean of θ_j not the same for players with no attempts?
3. For which players is the posterior mean closest to the observed free throw percentage? Why?
4. Which players have the smallest posterior SDs? Why?
5. Find and interpret 50%, 80%, and 98% posterior credible intervals for Joel Embiid.
6. Find and interpret 50%, 80%, and 98% posterior credible intervals for Kevin Ross.

24.1 Notes

24.1.1 Sample data